



**VietNam National University
University of Engineering and Technology**

EMBEDDED SYSTEM FUNDAMENTALS

(ELT3240, NHẬP MÔN HỆ THỐNG NHÚNG)

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Lecture 2:

Introduction to Embedded System Design

Objectives

In this lecture you will be introduced to:

- The embedded system design process.

Outline



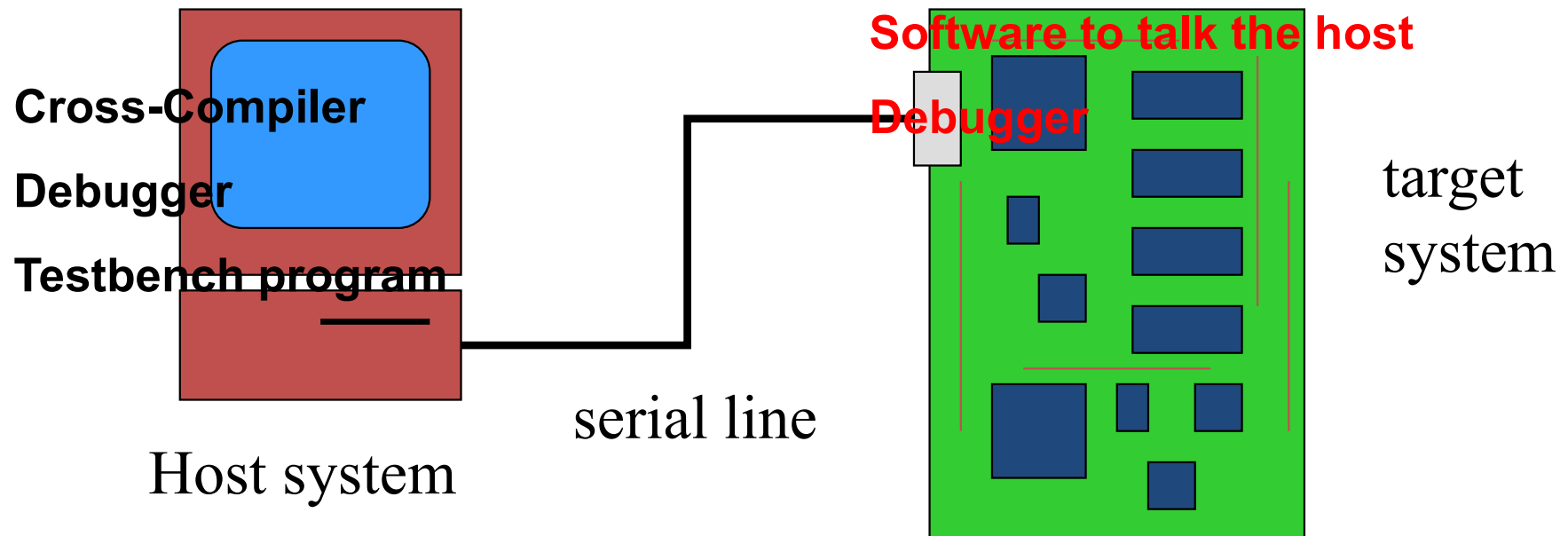
- **Embedded System Development with Platform**
- Microcontroller-based Embedded System Design
- Summary



What is necessary to develop embedded systems?

Development Environments

- Host system: a computer running programming tools for development.
Use a host system to
 - prepare software for target system
 - load programs into the target;
 - start and stop program execution on the target; and
 - examine memory and CPU registers
- Target System: the HW on which code will run

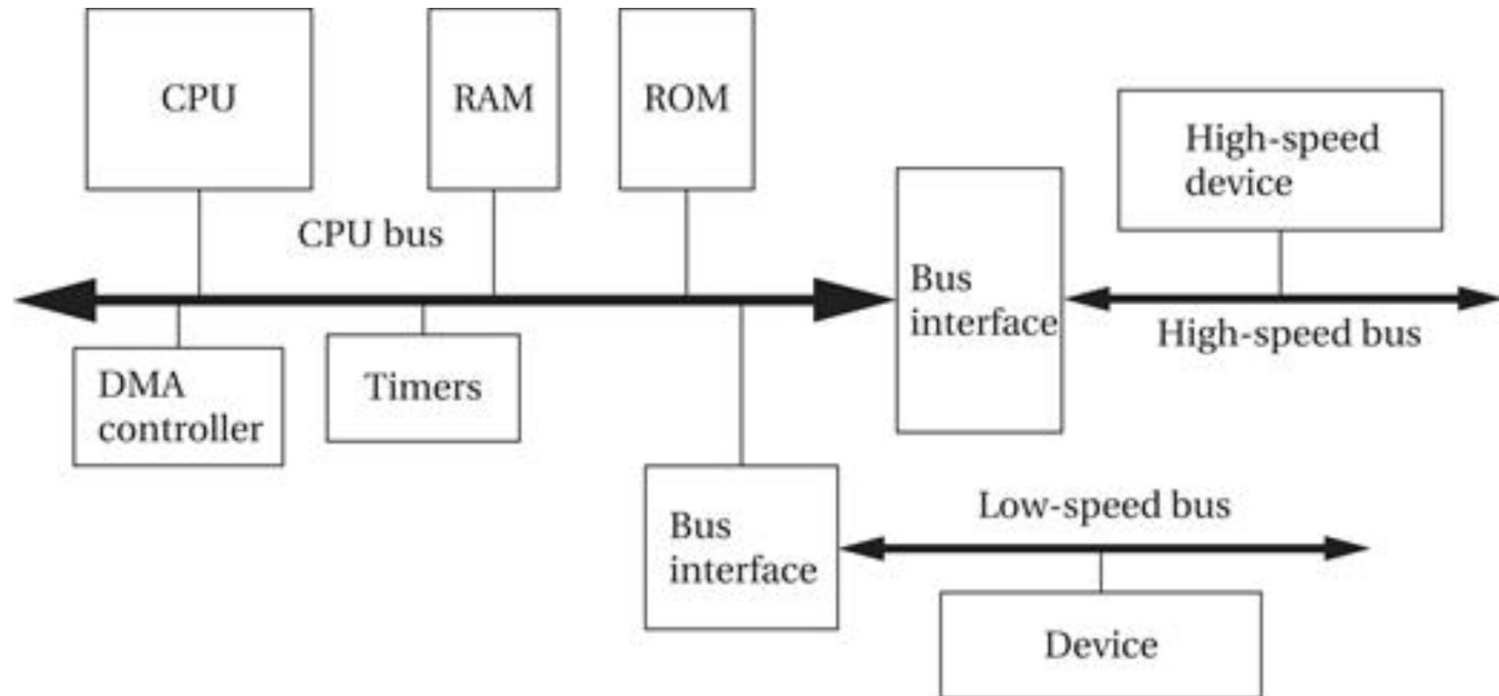


Designing with Platform



- **Platform** provides the environment in which we can develop our embedded applications quickly.
- Before real hardware is built, software can be developed and tested using platform
- It encompasses both **hardware** and **software** components

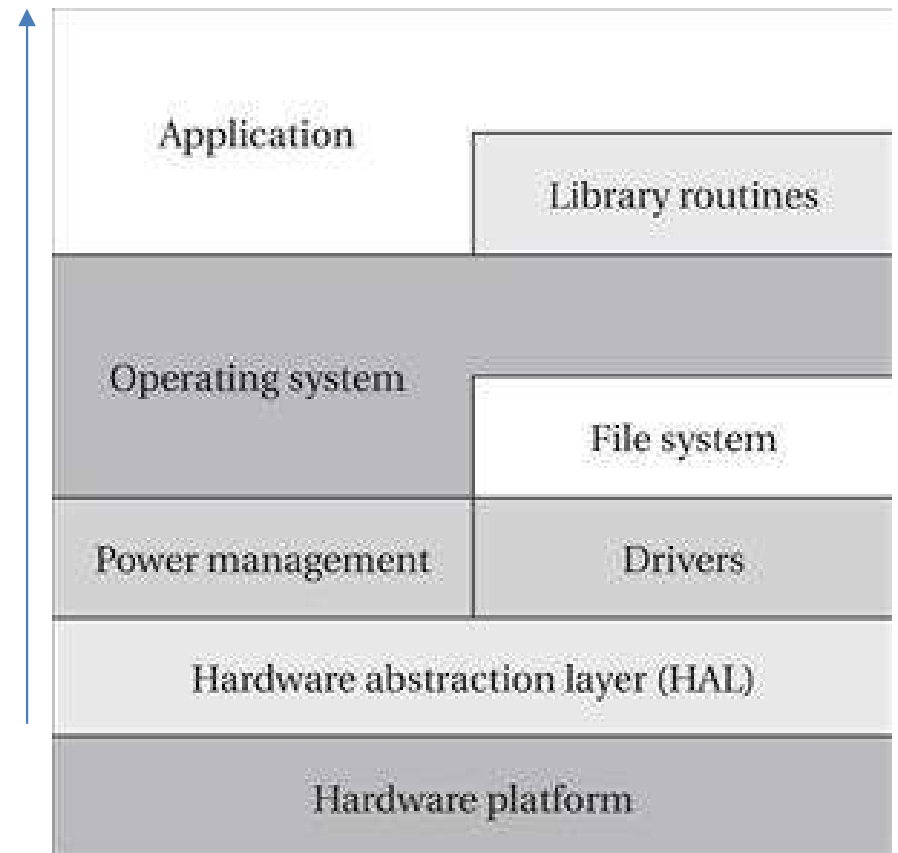
Designing with Platform: Hardware



- DMA provides direct memory access.
- Timers used by OS, devices.
- Multiple busses connect CPU and memory to devices.
 - low-speed: provides an inexpensive way to connect simpler devices
 - high-speed: allows fast devices to communicate efficiently
- Multiple buses serve two purposes:
 - reduce the overall load and increase the utilization of the buses
 - low-speed buses provide simpler and cheaper interfaces than high-speed buses

Designing with Platform: Software

- Hardware vendors generally provide a software platform to encourage use of their hardware.
- Software Platform provides core functions, utilities.
- Low-level functions depend on architecture--interrupt vectors, etc.

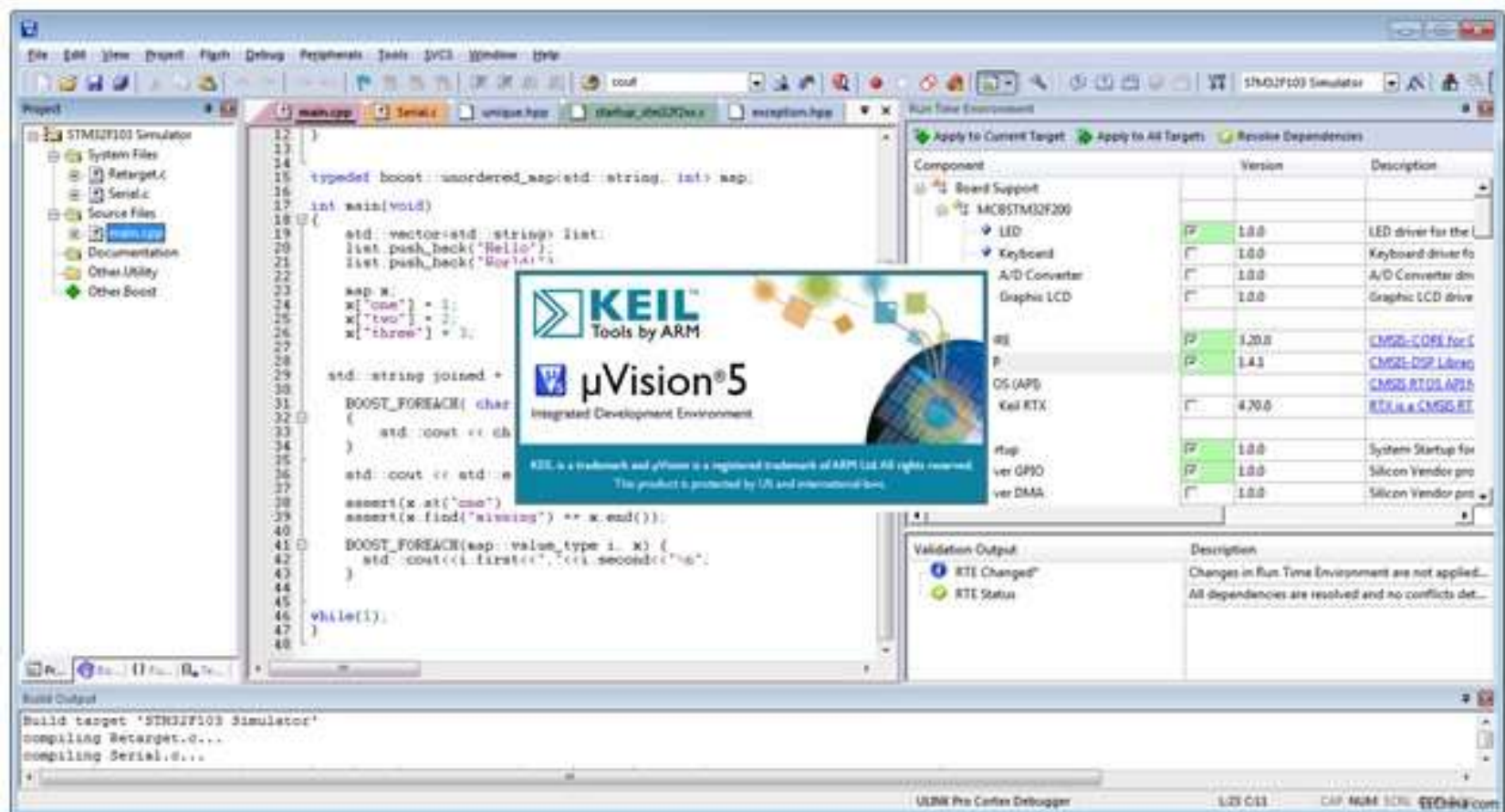


Software layers for an embedded system

Designing with Platform

What is needed to start (1):

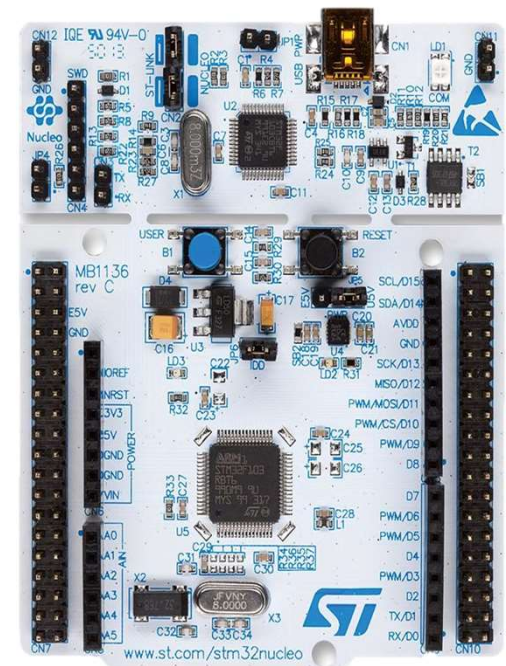
- Development tool suites:
 - STM32CubeIDE
 - Keil ARM-MDK



Designing with Platform

What is needed to start (2):

- Development board:
 - Development boards usually have the same CPU as the end product and provide many IO peripherals for the developed software to use as if it were running on the real end product
 - Designed by CPU manufacturer or others.
 - Includes CPU, memory, some I/O devices.
 - May include prototyping section.
 - CPU manufacturer often gives out reference design---can be used as starting point for your custom board design.



Designing with Platform

What is needed to start (3):

- Debug Adapter
 - To download and debug your program code on the microcontroller
 - Debug adaptor products vs. built-in USB debug adaptor

Designing with Platform

What is needed to start (4):

- Hardware Abstraction Layer (HAL)
 - Definitions of peripheral registers
- Device Driver
 - Functions for configuring and accessing the peripherals
- Documents and other resources
 - some example code
 - user manuals, application notes

Choosing a platform

Hardware

- CPU: choice of instruction sets, features (e.g. clock speed, bus data width, integrated peripherals), etc.
- Bus determines available I/O devices, system performance.
 - To pay attention to the required data bandwidths to be sure that the bus can handle the traffic
- Memory: **size** (for data and code), **speed** (determining system performance), **cost**.
- I/O devices vary in performance, cost.

Choosing a platform

Software

- Software Support:
 - Run-time components: code libraries, OS, File System,...
 - Development Environment, debugging tools, ...

Intellectual property

- Intellectual property (IP) is something that we can own but not touch
 - Hardware designs, source or object code, schematics, netlists, etc.
- Used at all levels of design:
 - Schematics for hardware reference design.
 - Drivers and run-time libraries.
 - Software development environments.

Debugging Techniques

How to validate code

- Run on host system.
 - Run in instruction-level simulator.
 - Run on cycle-accurate simulator.
- Run in hardware/software co-simulation environment.
 - In-circuit emulator: is a **specialized hardware tool** that can **help debug software in a working embedded system**
- Run on target system.

Breakpoints

- Breakpoint is the important debugging tool that specify an address at which the program's execution is to break.
- A breakpoint allows the user to stop execution, examine system state, and change state.
- Replace the breakpointed instruction with a subroutine call to the monitor program.

ARM breakpoints

0x400 MUL r4,r6,r6

0x404 ADD r2,r2,r4

0x408 ADD r0,r0,#1

0x40c B loop

0x400 MUL r4,r6,r6

0x404 ADD r2,r2,r4

0x408 ADD r0,r0,#1

0x40c BL bkpoint

uninstrumented code

code with breakpoint

- Save registers.
- Allow user to examine machine.
- Before returning, restore system state.

LEDs (Light-Emitting Diodes)

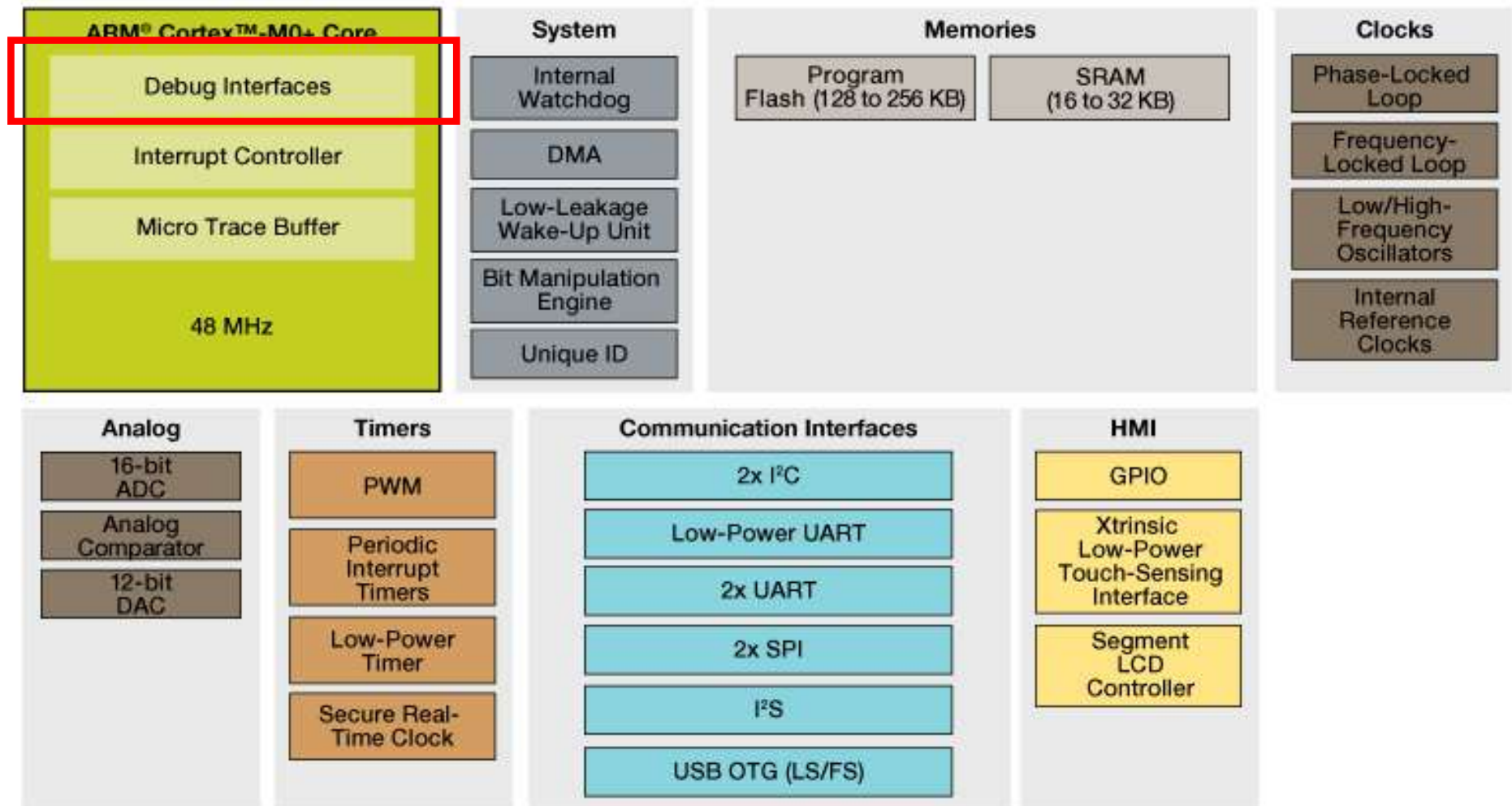
- Used to indicate the system state (such as error condition, idle state, ...) even if they will not normally be used in applications.

In-circuit emulators

- A microprocessor in-circuit emulator is a specially-instrumented microprocessor.
- Allows you to stop execution, examine CPU state, modify registers **in a working embedded system**.

In-circuit emulators

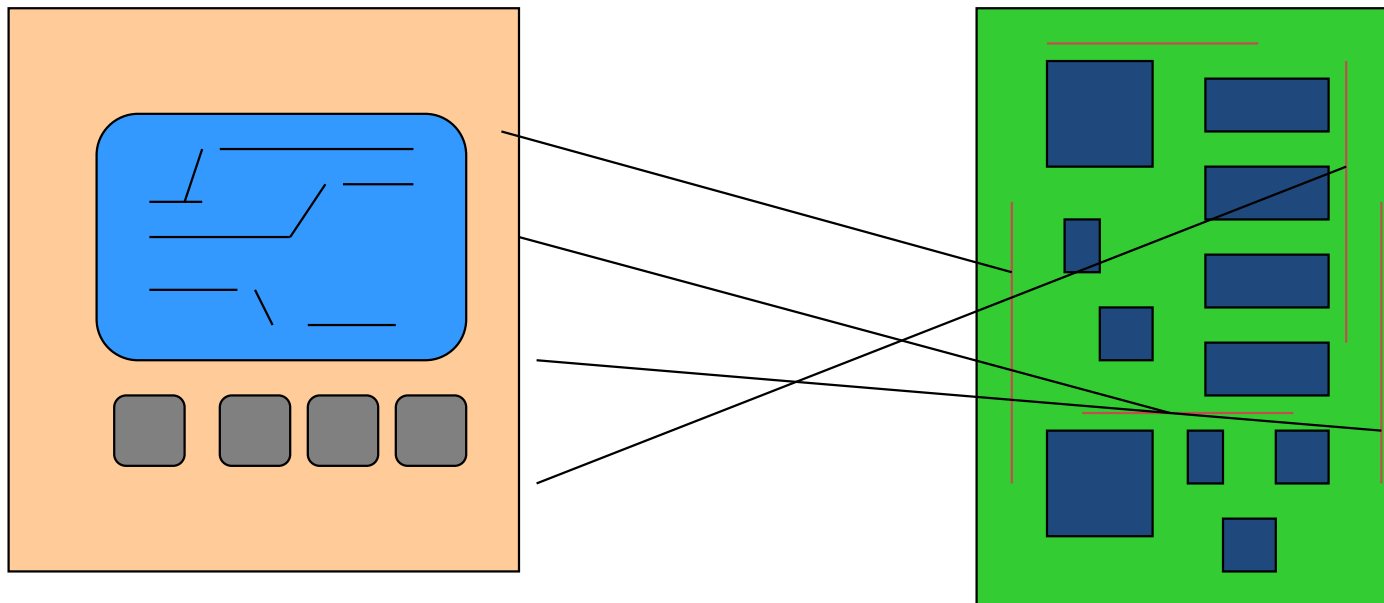
KL4x Family Block Diagram



☐ Standard ☐ Optional

Logic analyzers

- A logic analyzer is an array of low-grade oscilloscopes:
 - can sample many different signals simultaneously
 - Two sampling modes:
 - Timing mode uses an internal clock that is fast enough to take several samples per clock period in a typical system→(greater resolution, detecting glitches)
 - State mode, on the other hand, uses the system's own clock to control sampling, so it samples each signal only once per clock cycle.



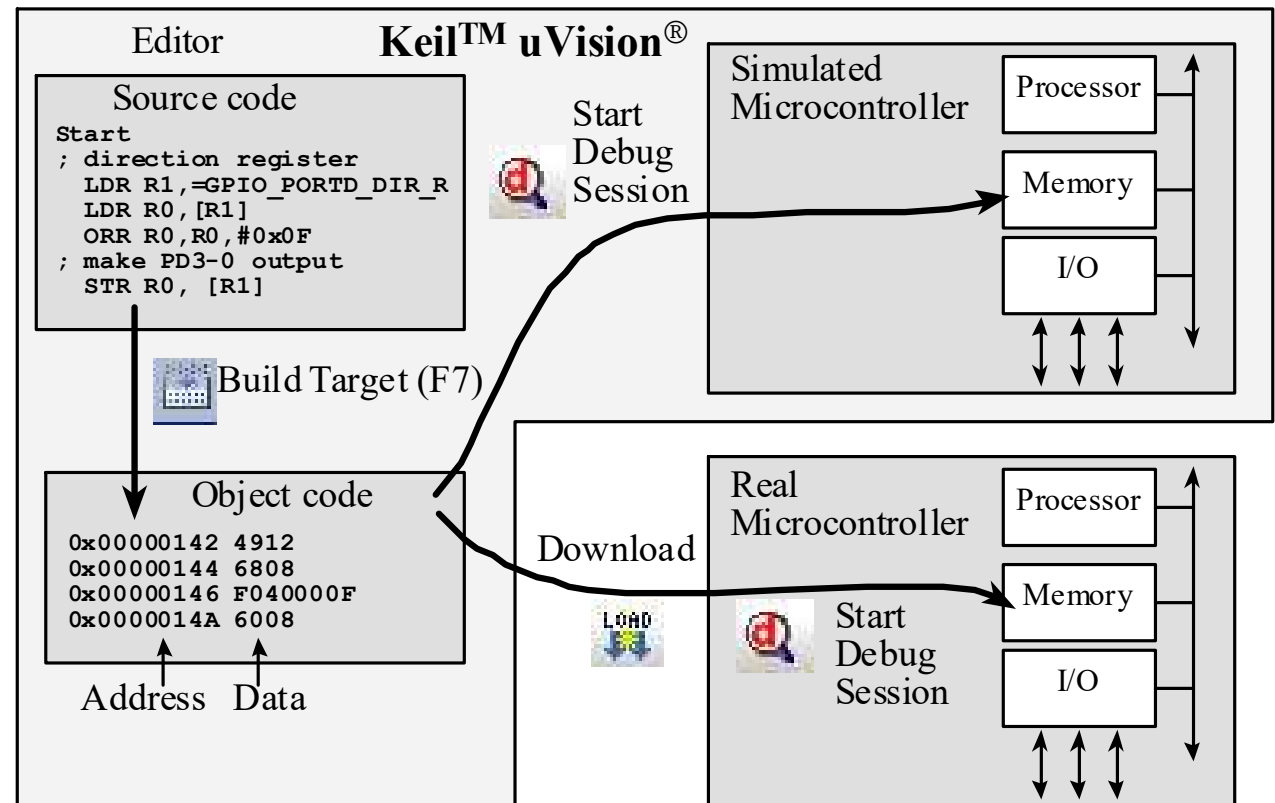
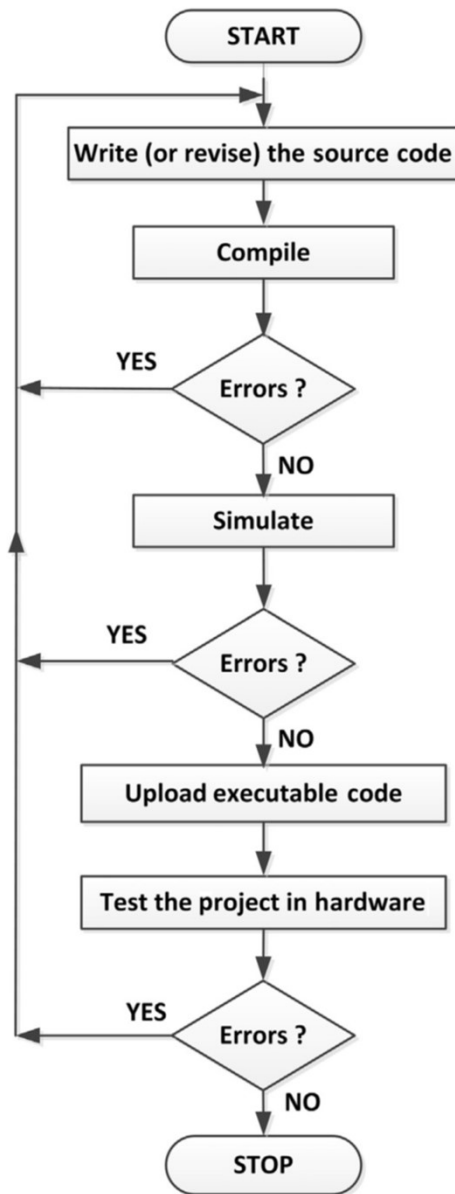
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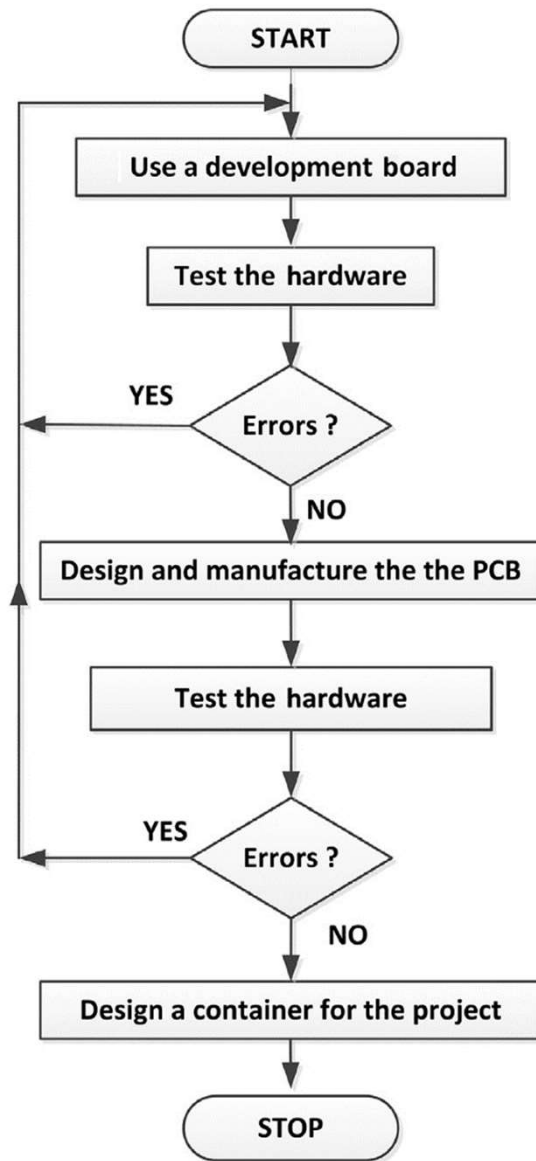
Microcontroller-based project development



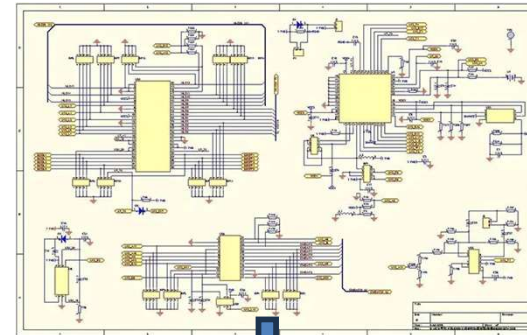
SW Development Environment

The Program Development Cycle

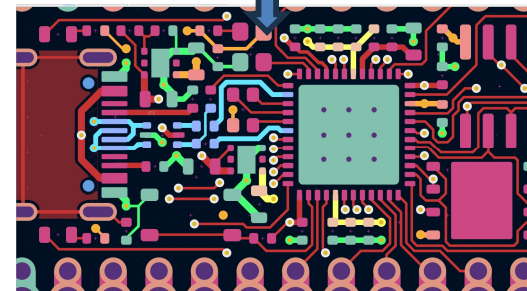
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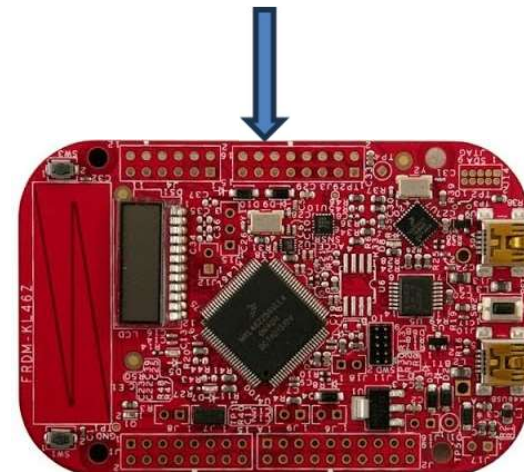
Program development cycle



Schematic



PCB Layout



PCB Board

Outline

- Embedded System Development with Platform
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- **Summary**

Summary

- **What we Learned**

- Basic concepts of Embedded System-on-Chips
- Necessary things to develop embedded systems
- The embedded system design process and different abstract levels for describing Embedded Computing Systems